EIGSEP Memo 007

Marjum Pass 2026-07: the self-EMI tests, step by step

EIGSEP Team

Living document. This version: 2026-10-04

Abstract

On 07-14 and 07-15, with box-air on a workbench about 15 m from box-gnd, we changed box-air's electronics one piece at a time and watched the correlator: the Panda's power and its supply filters, the LIDAR, both IMUs, the temperature-controlled load, the CU, the RF switch and its filters, and the motors. Every test ran while the correlator recorded, so each configuration is in the data files. This memo documents the four sessions (90 steps, 9 of them changes seen in the data with no note), places every step in time, and extracts the mean auto- and cross-spectra of each configuration and the change at each step. It is the record that the next version will use to infer what each component emits. Every figure, table and quoted number is generated by `make_products.py` in this directory (Appendix C).

1 Summary

- **Every step is placed in the data.** The notes are timed by dashboard screenshots whose file names are PDT (UTC−7), not site time (Appendix B). Each step's change is then found in the data: from the sensor metadata where a sensor was plugged or unplugged, by a fit to the spectra between screenshots, or at the end of a box-air dropout (§3).
- **Input 4 was box-air in phase B**, input 3 box-gnd: switching the Panda off removes box-air's excess on input 4 and barely touches input 3 (§2). Memo 001 now takes this mapping from here.
- **The extracted data**: the mean box-air, box-gnd and cross spectrum of every step, with standard errors and half-interval means, are in `tables/segments.npz`; times, intervals and summary changes are in `tables/steps.json` (§3.4).
- **First look, not yet interpretation**: the Panda adds 21.1–228.8% to box-air's Panda-off power over 150–225 MHz in all 16 on/off comparisons, with narrow features, and changes the cross. Physically plugging or unplugging the LIDAR or an IMU changes box-air across most of the band. The steps whose notes repeat the previous configuration change at most 2.5% of box-air channels by more than three times the drift (§3.3).

2 The tests and the data

Setup. Box-air (the raised science dipole with its electronics box: the Panda computer, motor controllers, IMU, LIDAR) sat on a portable workbench about 15 m from box-gnd (the static ground dipole) for all four sessions, before the lift to ∼30 m on 07-15 (memo 001). "CU" in Aaron's session-3 notes ("cu off", "cu on") is either the 6 V battery that powers the RF electronics or the copper tape used for EMI shielding (Aaron, 2026-10-04); the data constrain which (§4.3). "IMU" in Aaron's notes is the external azimuth IMU (`imu_az`); Christian's notes name the box (elevation) IMU separately. Table 1 lists the sessions. The field notes for 07-14 also record gasketing the Panda's power connector and RJ45 ports and adding BLP-5 filters and a capacitor on the Panda power input (memo 001 §Timeline); session 1 is that work.

Records. Three independent records time the tests, and all are used:

Table 1: The sessions. Times (PDT; UTC = PDT+7 h) span the first change to the end of the last averaging interval; the files are the first and last correlator files `corr_<name>.h5` (named by close time, UTC) with data in that span. Inputs are SNAP inputs. Session 2's stored cross `35` is not valid: the multiplexer copy of input 4 was off. Integrations are 0.268 s in sessions 1–3 and 0.537 s in session 4.

session	by	PDT	box-air	box-gnd	cross	first file	last file
S1	Aaron	07-14 08:46–09:39	4	3	35	`20260714_154814Z`	`20260714_164225Z`
S2	Christian	07-14 15:50–16:18	4	3	35	`20260714_225036Z`	`20260714_231831Z`
S3	Aaron, Christian	07-14 19:19–19:45	4	0	04	`20260715_021939Z`	`20260715_024526Z`
S4	Aaron	07-15 10:00–11:04	4	0	04	`20260715_170239Z`	`20260715_180456Z`

1. *Notes.* Aaron's `obs_notes.txt` (sessions 1, 3, 4) gives, for each screenshot, the configuration after the change. Christian's notes (`chb_notes.txt`, session 2) give 25 steps with no times. Both are transcribed into `tests.json`, which keeps the field text verbatim beside a restated configuration.
2. *Screenshots* of the live correlator dashboard (119 files), whose file names carry the laptop's clock. They and the notes are in `marjum-2026-07/imgs/self_emi_testing/`.
3. *Sensor metadata* in every correlator file (`metadata/imu_az`, `imu_el`, `lidar`, `tempctrl_load`, ...): one status record per integration, which turns to `error` when a sensor stops answering and back to `update` when it answers again. It times physical plugging and unplugging without reference to any clock but the data's, and it stops entirely when the Panda is off.

Inputs. Sessions 1 and 2 are in wiring phase B (memo 001), with the two antennas on inputs 3 and 4 and the cross between them available only as `35`, through the multiplexer copy of input 4 on input 5. Which antenna was on which input in phase B was open in memo 001 (the RFI pipeline's antenna policy, `curation/antenna_resolution.json`, assumed 3 = box-gnd). The Panda tests settle it: in session 1, switching the Panda off removes (18–41) $\times 10^3$ counts from input 4, averaged over 150–225 MHz (21–47% of its Panda-off power), and changes input 3 by at most 6×10^3 counts (Table 2). An intervention that acts on box-air changes input 4, so input 4 is box-air and input 3 is box-gnd; Christian, testing box-air's sensors in session 2, watched input 4. The multiplexer copy was off for all of session 2 (memo 001), so that session has no valid cross. Sessions 3 and 4 are in phase C: box-gnd on input 0, box-air on input 4, cross `04`. Integration time doubled before session 4.

Clock. The screenshots' file names are the field laptop's clock, which ran on PDT (UTC−7), an hour behind site time; the dashboard shows data 8.6–16.6 s old. Both are fixed by three screenshots of the 07-17 transmitter on/off test (Appendix B). Read as MDT, the notes land an hour early, on unrelated data.

3 Method

3.1 Change times

A step statistic $S(t)$ compares, channel by channel over 25–240 MHz, the mean log power of all inputs (and |cross| where valid) over the next 4 s or so with the previous, in units of that difference's robust scatter within ±120 s, averaged over channels. Dead integrations and *dropouts* (box-air's band level more than 10 dB down, typically while the box was being rewired) are left out. Step k's change must lie between its previous screenshot less 24.0 s and its own less 7.0 s, and one change per step is chosen jointly for the session to maximize the summed S. Three sources override this fit: the sensor metadata, where a note names a sensor that was plugged, unplugged or switched off (29 steps, listed in Appendix B; Panda power-offs are timed by the end of all metadata); the end of a dropout inside the step's window (8); and, where the fit is visibly wrong on full-resolution waterfalls, a time set by eye with its reason in `tests.json` (11). Of the 9 unlogged entries, four are screenshots with no note and five are changes found in the data: a Panda power-off (S1-u1), the Panda's return before the next note (S1-u2, S2-u0, S4-u0), and a fall in box-air at 150–225 MHz (S4-u4). A step whose S is under four times its local median is *weak* (22 steps): no clear change, so its boundary is uncertain. Where the metadata and the spectra disagree on a change by under 60 s (session 2), the span between them is left out of both intervals.

Christian's 25 steps carry no times. Taking the k-th screenshot as step k fits the metadata except for steps 9–16, where the LIDAR is off continuously through three screenshots; there the metadata fit a mapping in which steps 9 and 16 (both "same as 1") have no screenshot and three screenshots show step 10. `tests.json` records it.

3.2 Averaging

Step k's interval runs from its change plus 2 s to the next change less 2 s. Within it, integrations in a dropout are dropped, and so are integrations whose box-air median level over 25–240 MHz departs from the interval's median by more than 1 dB (boot flashes, brief losses). Samples beyond six robust standard deviations of their channel's median are flagged. These order statistics only flag. Every reported spectrum is the mean of the unflagged samples, with its standard error from their scatter, and the means of the interval's two halves.

3.3 The change at a step and its null

Each step k (A) is compared with the step before it (B). Contributions add, so the change is the difference A−B: in counts for each auto-correlation, and the complex difference for the cross. Figures show A, B and |A−B| on one logarithmic axis. The thermal standard error is too small a yardstick. Over minutes the sky and the gains drift, and nearly every step, including those where nothing changed, differs from its predecessor by many standard errors. The null used instead is the difference between the two halves of each of the two intervals, combined in quadrature. It carries the drift and the intermittent RFI the steps carry, and for pure noise it is $\sqrt{2}$ times the expected change. A step's figure of merit is the percentage of channels (25–240 MHz, without the FM band) where the change exceeds three times this null, smoothed over 17 channels. The check can come out the other way: the three steps whose notes repeat the previous configuration (S1-14, S2-21, S4-47) change at most 2.5% of box-air channels and 3.9% of cross channels. Box-gnd is noisier against this null (up to 21.0% in the same controls), so box-gnd percentages below that mean nothing.

3.4 Products

- `tables/segments.npz`: for every step with an interval, keys `<id>/boxair`, `<id>/boxgnd`, `<id>/cross` (complex) with `_se`, `_flagfrac`, `_h1`, `_h2`, plus `<id>/interval_utc` and `<id>/n`; and `freq` in MHz. Units are raw correlator counts per integration.
- `tables/steps.json`: per step, the change time (PDT and UTC), its source and statistic, the interval and integrations kept, the file and integration index at the change, the configuration, and summary changes: band-mean A−B over 30–85 and 150–225 MHz (counts) and the percentage of channels beyond three times the null, quoted in the text below.
- Figures per session: waterfalls with the change times (`figures/session_S*.pdf`), and A, B and |A−B| at every step (`figures/steps_S*_p*.pdf`).

4 The sessions

Each session has a waterfall figure (all inputs, change times, sensor reports) and step figures: for every step, A (this step), B (the previous step) and |A−B| for box-air, box-gnd and the cross. Table 3 (Appendix A) gives every step's timing, file, integration index, configuration and field note. This section says what was done and what changed. What the changes mean is left to the next version of this memo.

4.1 S1: the Panda's supply filters (07-14 morning)

Aaron cycled the Panda's power six times while rearranging the filters on its supply: two BLP-5 low-pass filters around a capacitor with a choke (S1-02), without the choke (S1-03), the choke restored (S1-04); the capacitor and one BLP-5 (S1-06, with which the azimuth drive would not run); the capacitor alone (S1-07); the two BLP-5s moved to the 12 V switch branch (S1-08), removed (S1-10), on the 6 V line (S1-11, S1-13, S1-14), and removed again (S1-15). Two further power cycles precede filter changes: one at 09:02:41 that is not in the notes

(S1-u1), and S1-09 at 09:11:02, after which the Panda ran in an unrecorded configuration (S1-u2) until 09:14:58. Box-air dropped out for 235 s from 09:18:33 and for 56 s from 09:31:30 while the filters were moved.

Every Panda power-off removes the same excess from box-air (Fig. 17, Table 2). Of the filter changes, removing the choke (S1-03, 3% of channels) changes box-air no more than the repeat S1-14 (3%). Moving the BLP-5s does change it: 49% of channels at S1-08 and 33% at S1-15, in both cases lowering box-air by (15–20) $\times 10^3$ counts over 150–225 MHz. Box-gnd changes at three of these steps (S1-03, S1-08, S1-11), mostly inside a deep notch that input 3 shows at 110–150 MHz and that moves through the session (Fig. 1).

4.2 S2: the sensors, physical and software disconnects (07-14 afternoon)

Christian disconnected and reconnected the LIDAR and the azimuth IMU physically and in software, switched the box IMU off in software, turned the temperature controller on and off, and power-cycled the Panda. Box-air's input was dead until 15:44, dropped out 18 times in the session, mostly for a few seconds, and the multiplexer copy was off, so there is no cross. The physical LIDAR disconnects (S2-02, S2-17) remove (3.5–4.7) $\times 10^5$ counts from box-air over 150–225 MHz, 70–75% of its power there, and reconnecting it after the power cycle (S2-23) adds 6.3×10^5. The software disconnect of the azimuth IMU (S2-20) and its repeat (S2-21) change little (4% and 1% of channels); other software steps do change the spectrum, but in this session their times are the least certain (§3). At 16:17:25, after the last note, the box IMU and the LIDAR stopped answering, and box-air lost 6.2×10^5 counts over 150–225 MHz, 72% of its power there (S2-u1).

4.3 S3: sensors, the load, "cu" and the azimuth drive (07-14 evening)

With the LIDAR, the external IMU and the load all unplugged, Aaron (and Christian for S3-19b) plugged in the IMU, put it in standby and unplugged it; did the same with the LIDAR; switched "cu" off and on; plugged in the IMU with the load and unplugged both; and ran the azimuth drive for about ten seconds. The metadata time the IMU and the load to the second, and the spectra change at those times (Table 4). Plugging and unplugging the IMU change most of the band in box-air and the cross (71% and 90% of box-air channels). Standby changes neither the IMU's nor the LIDAR's contribution by much (7% and 12%), and "cu off" and "cu on" change nothing (0% and 3%): no dropout, box-air's 150–225 MHz mean within 7% and box-gnd's within 3%, and with "cu" off box-air still responds to the IMU and load (S3-25). If "cu" is the 6 V battery of the RF electronics, the receivers had another supply at the time; copper tape removed and restored fits more simply. The notes alone do not decide it. The load with the IMU adds 1.3×10^5 counts over 150–225 MHz, about doubling box-air there. The azimuth drive running and stopping changes box-air and the cross across a third to two thirds of the band.

4.4 S4: reattaching everything, the RF switch and the motors (07-15 morning)

Starting from the Panda with nothing attached, Aaron added the load, the LIDAR and the IMU one at a time, removed the IMU, then plugged in everything else. He power-cycled the Panda, attached the RF switch, its long cable, the azimuth potentiometer and the load, and tried zero, one and two BLP-5s on the RF switch's 6 V line and two on its 12 V line, the note reading "think that rf switch w/o blp-5 filters is a problem". He then unplugged and replugged the RF switch, plugged in the elevation and azimuth motors, the LIDAR and the IMU, put the IMU in standby, removed it with the RF switch, and power-cycled the Panda twice. Box-air dropped out four times while the RF switch's filters were changed.

Plugging in "everything else" (S4-36) adds 3.9×10^5 counts to box-air over 150–225 MHz, tripling it there, the largest single addition of the four sessions. The IMU again changes most of the band when plugged and unplugged (73% each), and putting it in standby (S4-50) does too (82%), unlike in session 3. The RF-switch filter steps change box-air in 20–26% of channels; the replugged RF switch (S4-42) adds 5.9×10^4 counts at 150–225 MHz. The first Panda power-off of the pair (S4-52) is not resolved: its off state lasts at most 13 s before the boot flash, and its interval shows no change.

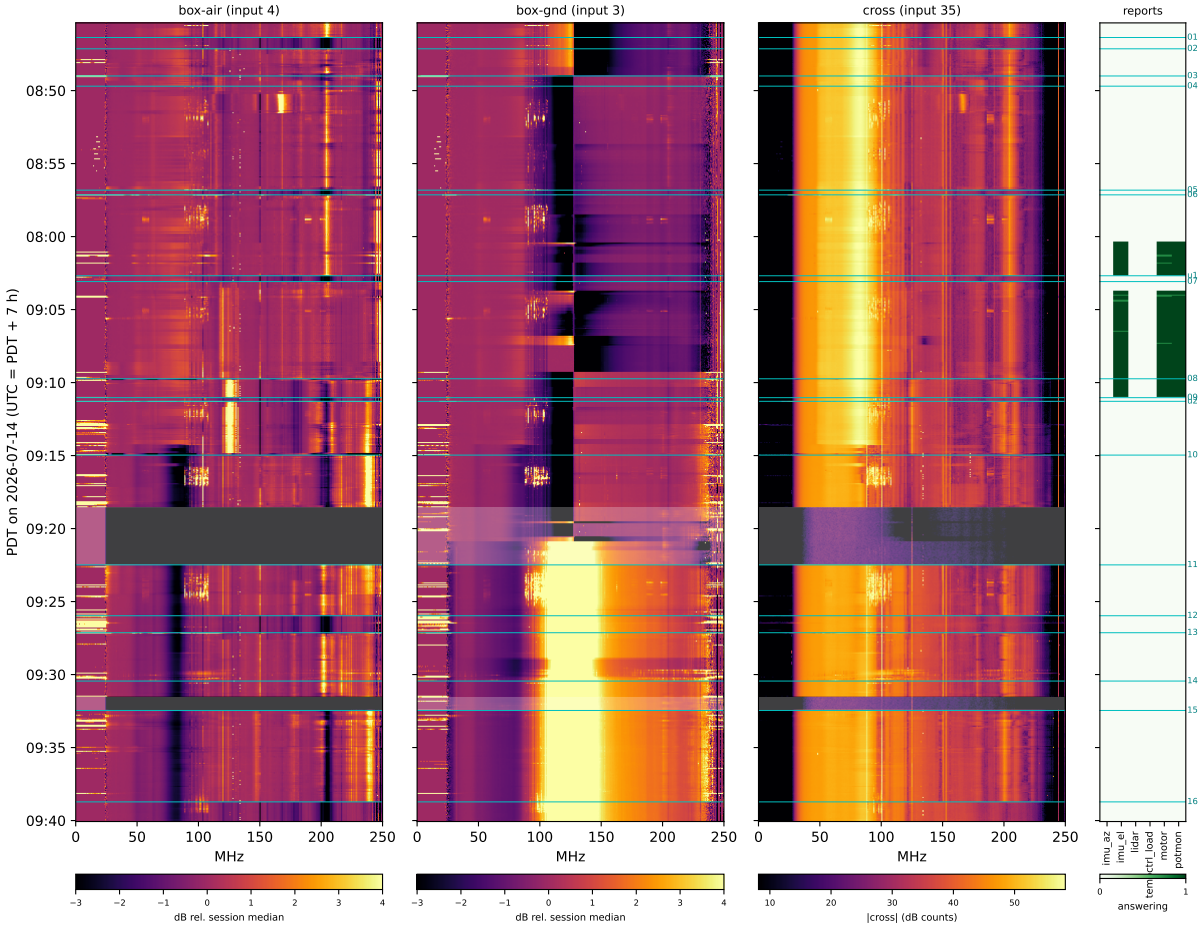


Figure 1: Session 1. Box-air and box-gnd in dB relative to the session median per channel; |cross| in dB counts; sensor reports (green: answering). Cyan lines: change times; pale bands: box-air dropouts. Labels at right are step ids.

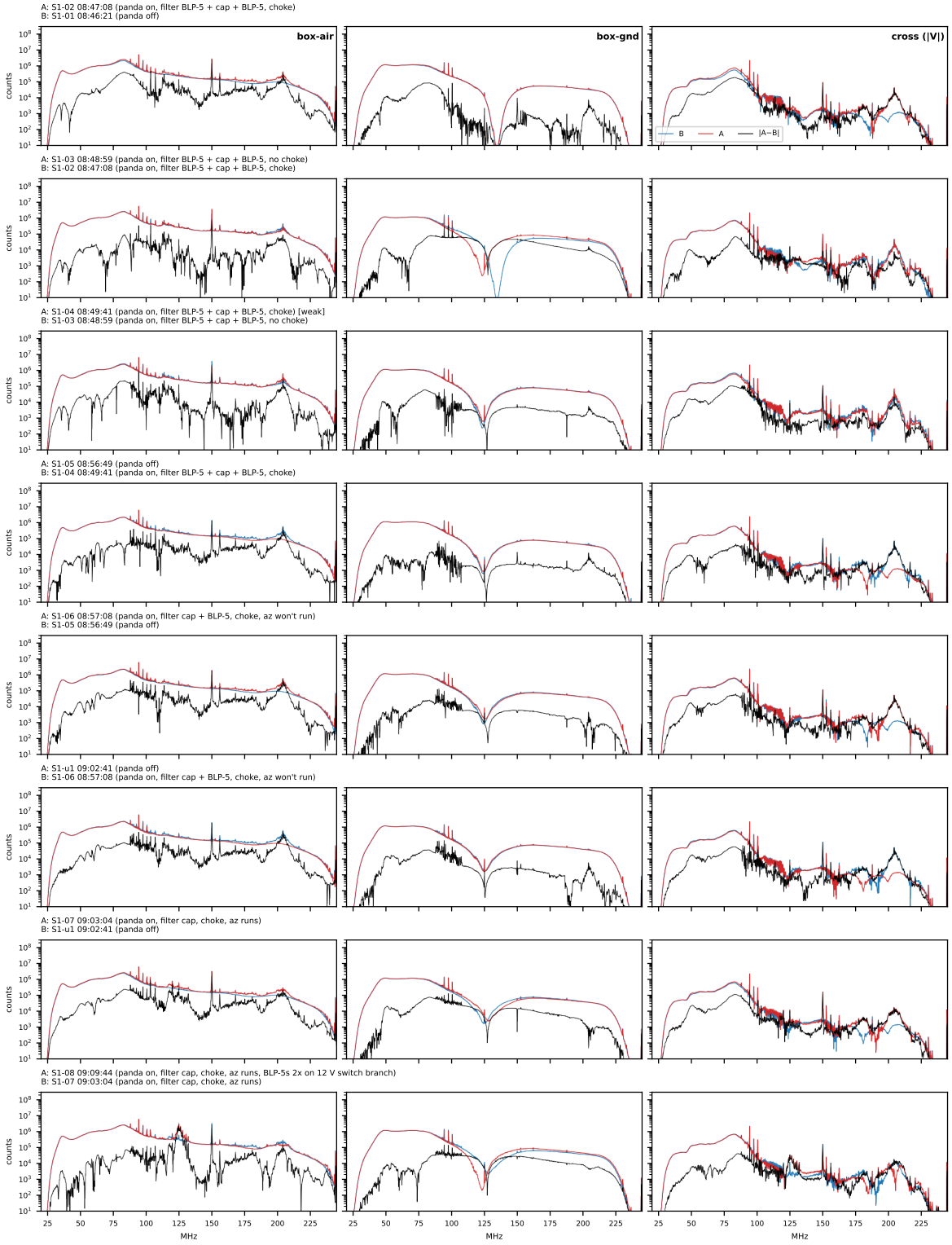


Figure 2: Session 1: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 1 of 3)

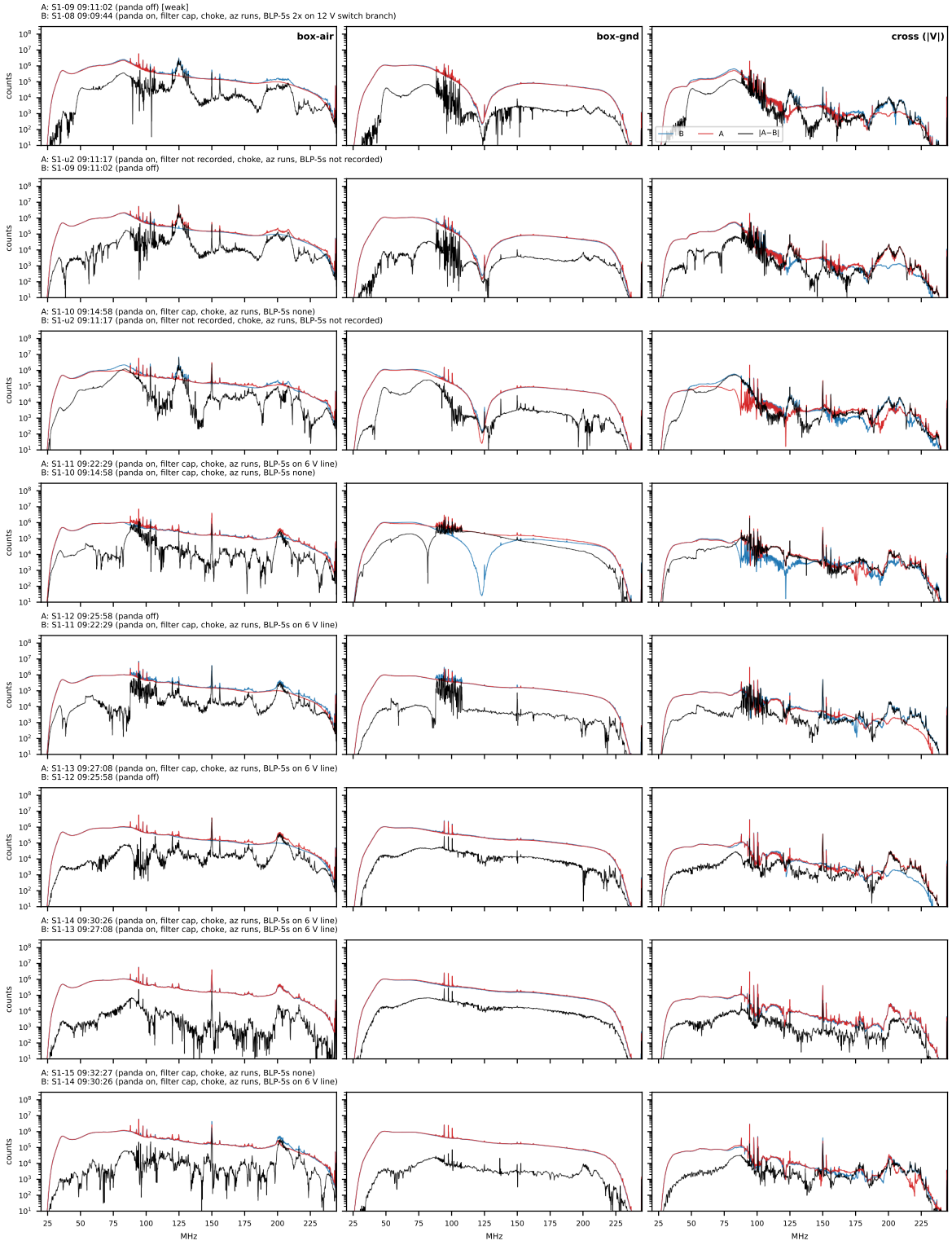


Figure 3: Session 1: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 2 of 3)

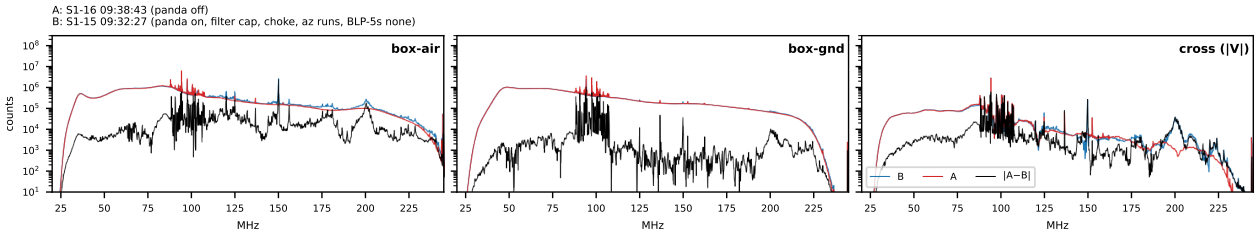


Figure 4: Session 1: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 3 of 3)

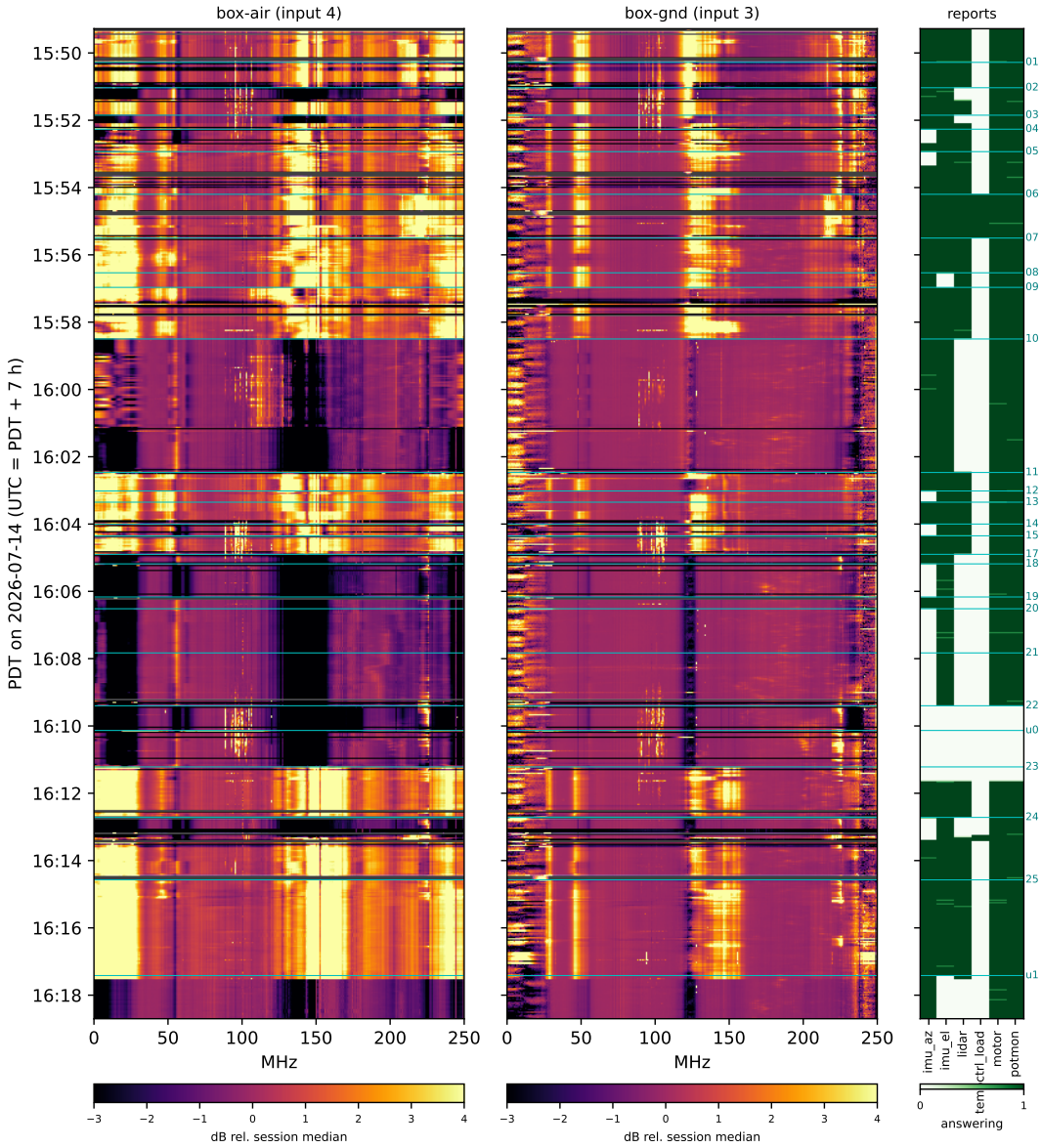


Figure 5: Session 2, as Fig. 1. No cross: the multiplexer copy was off.

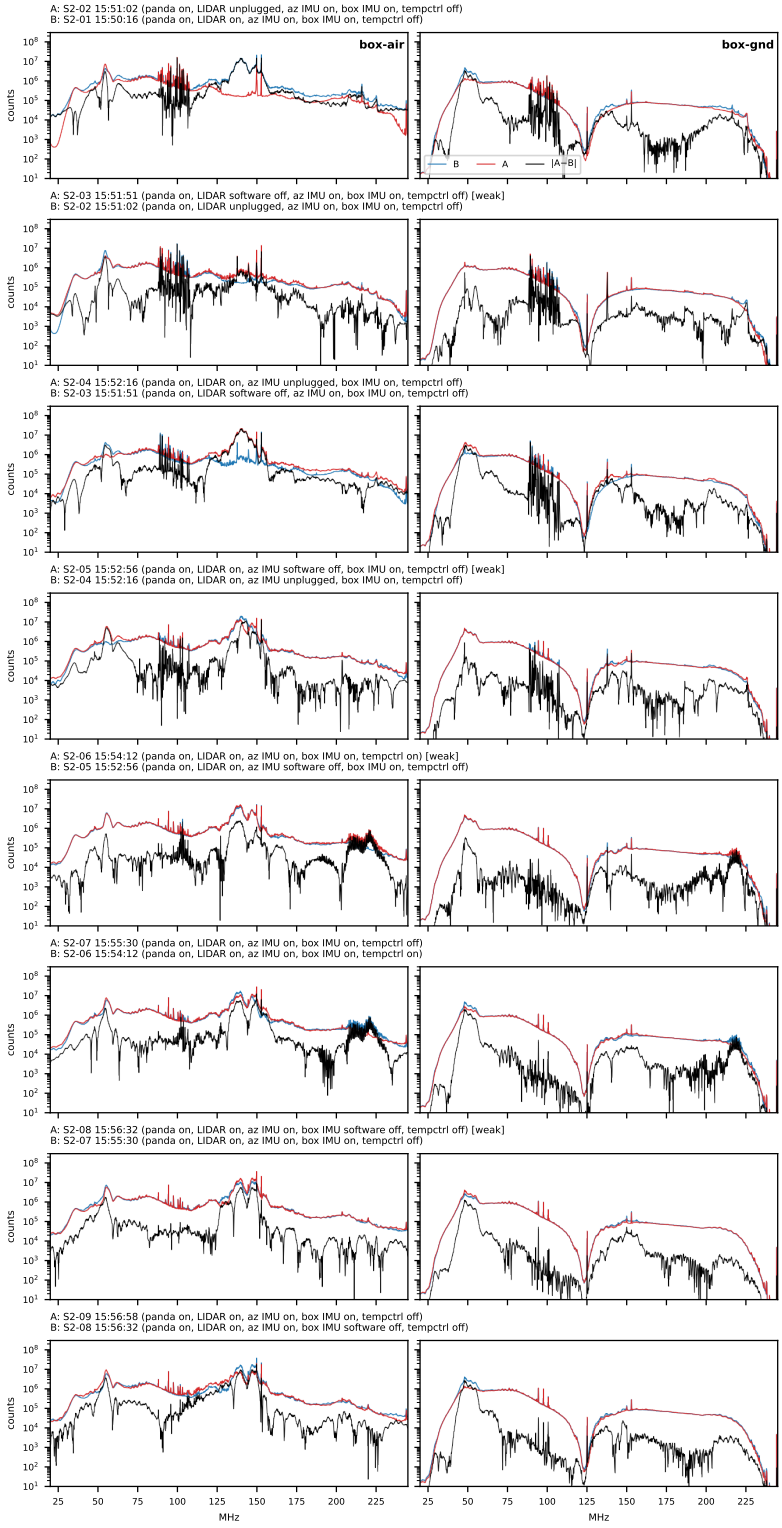


Figure 6: Session 2: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 1 of 3)

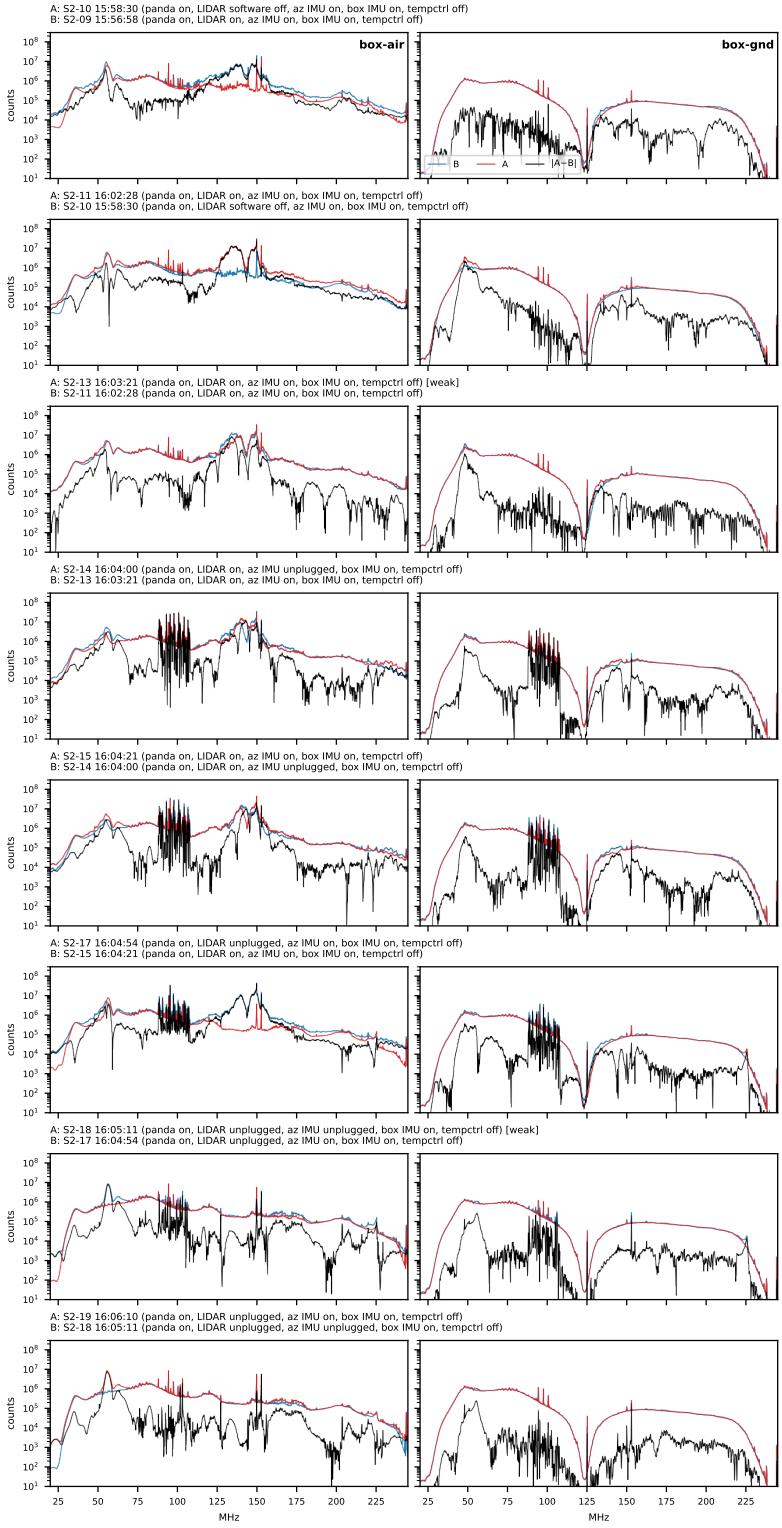


Figure 7: Session 2: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 2 of 3)

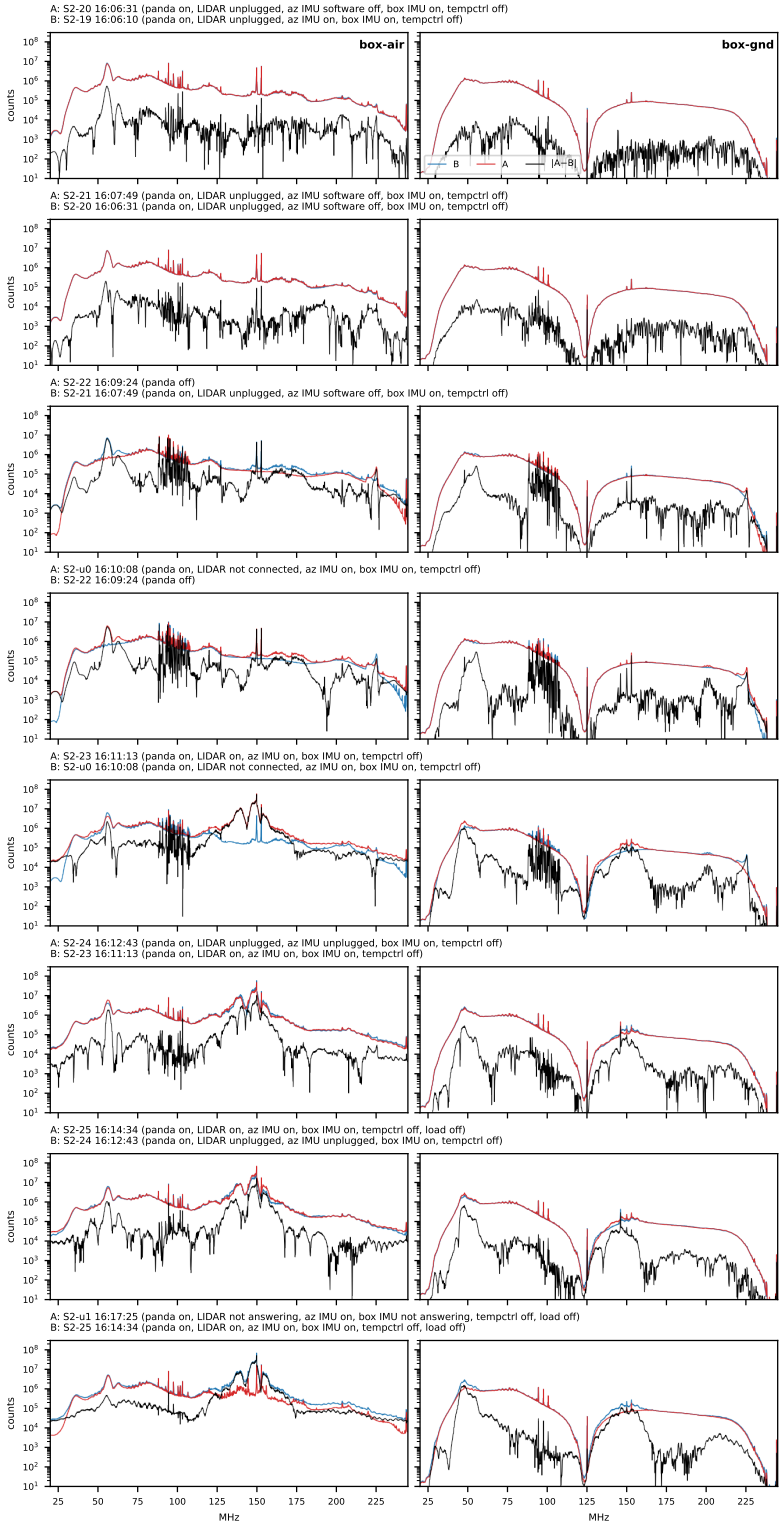


Figure 8: Session 2: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 3 of 3)

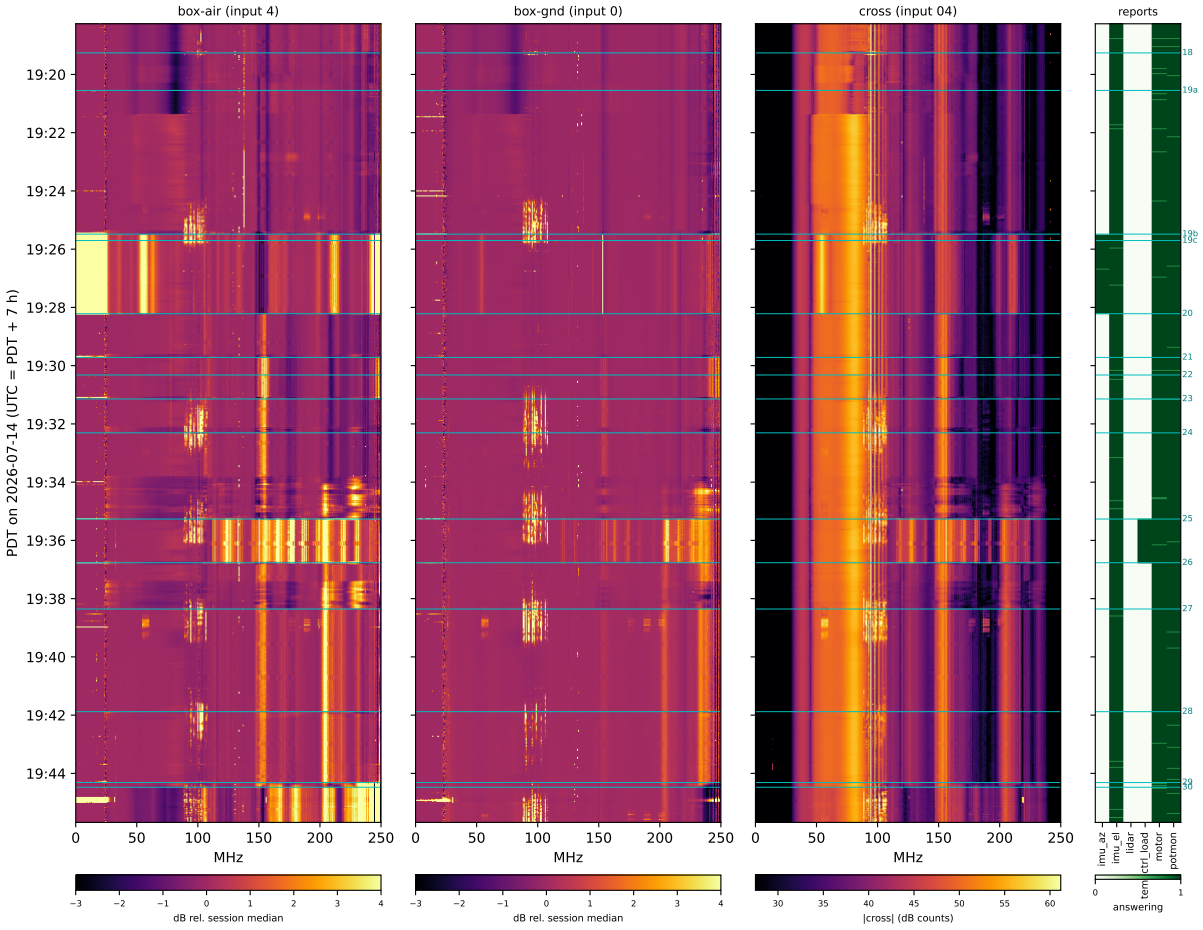


Figure 9: Session 3, as Fig. 1.

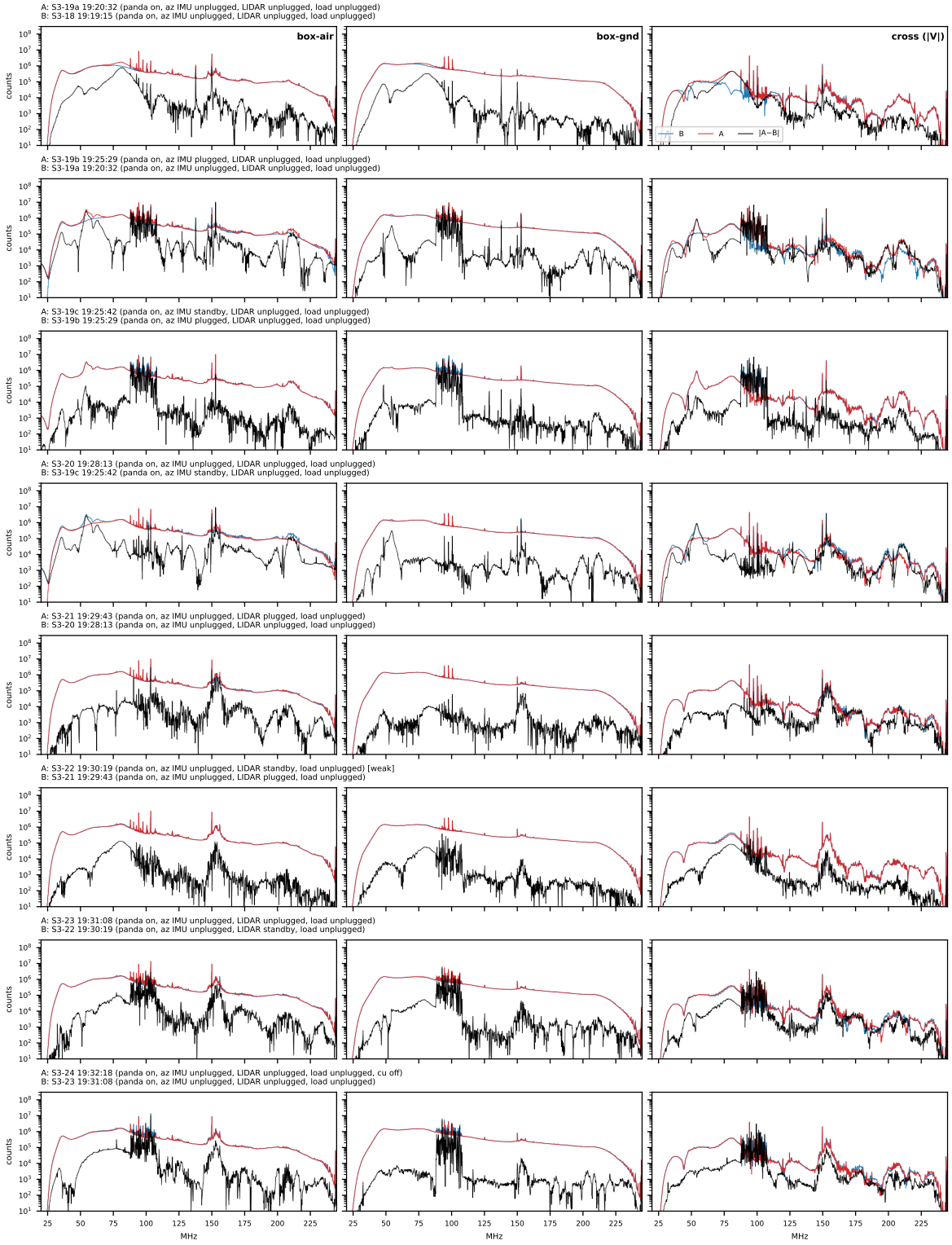


Figure 10: Session 3: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 1 of 2)

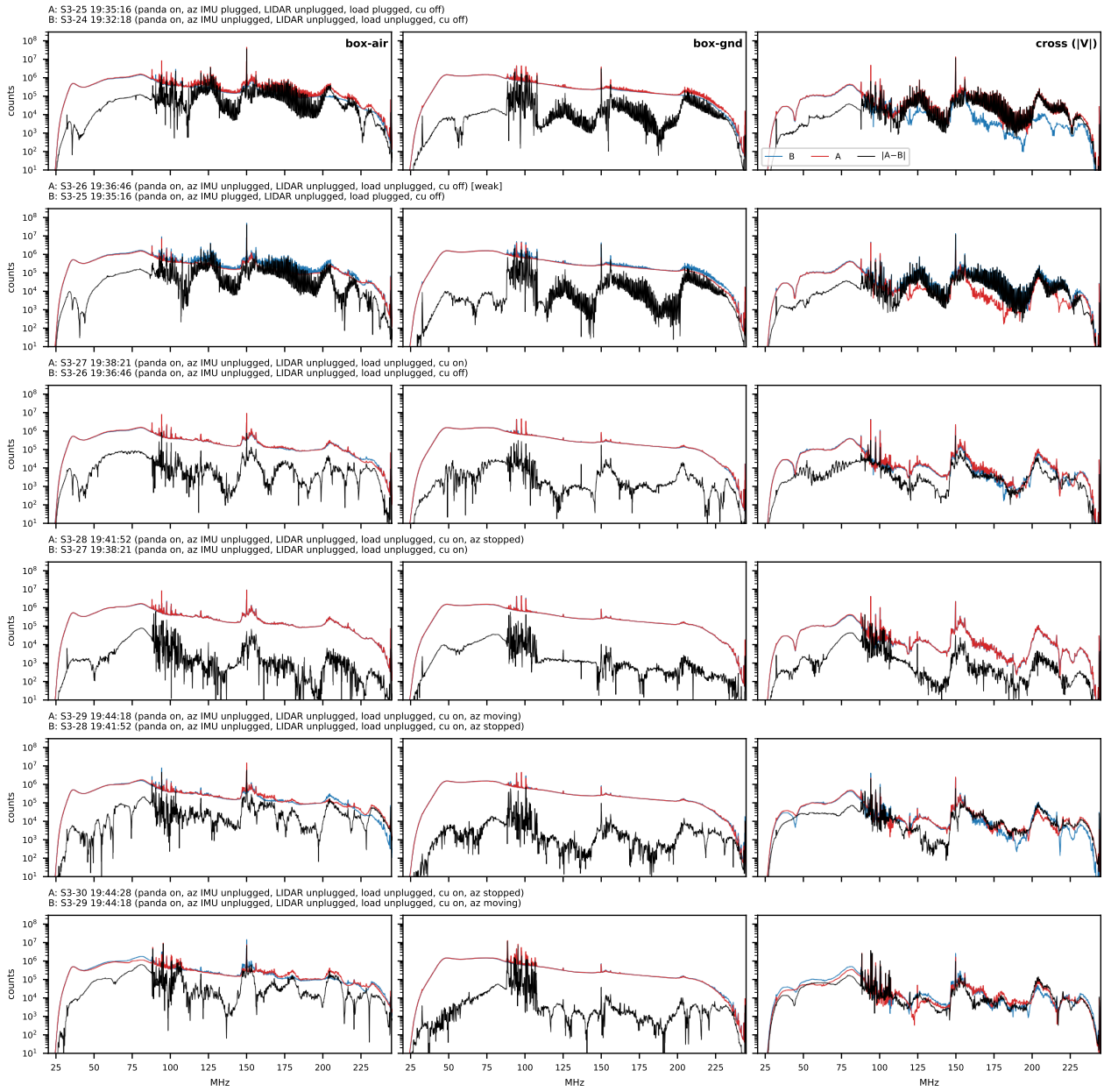


Figure 11: Session 3: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 2 of 2)

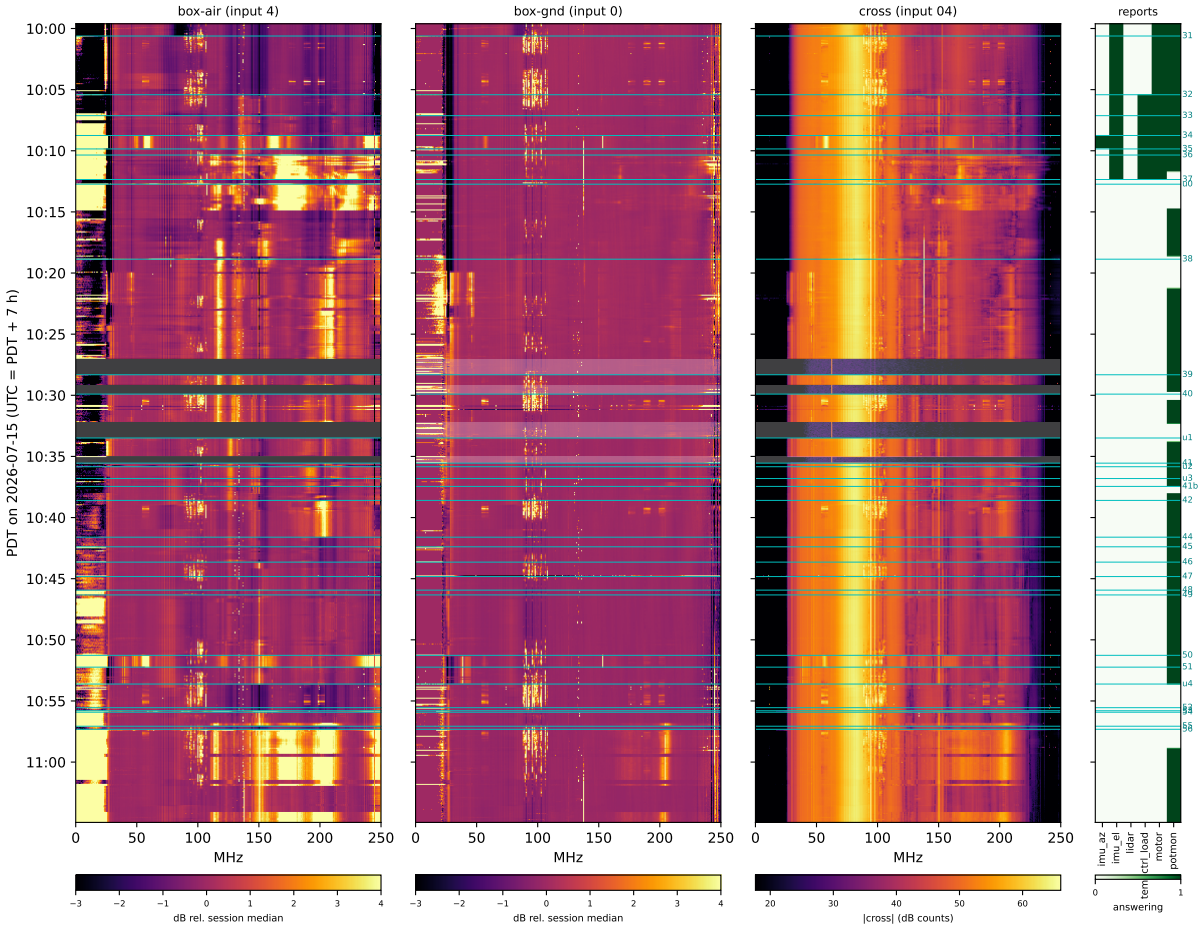


Figure 12: Session 4, as Fig. 1.

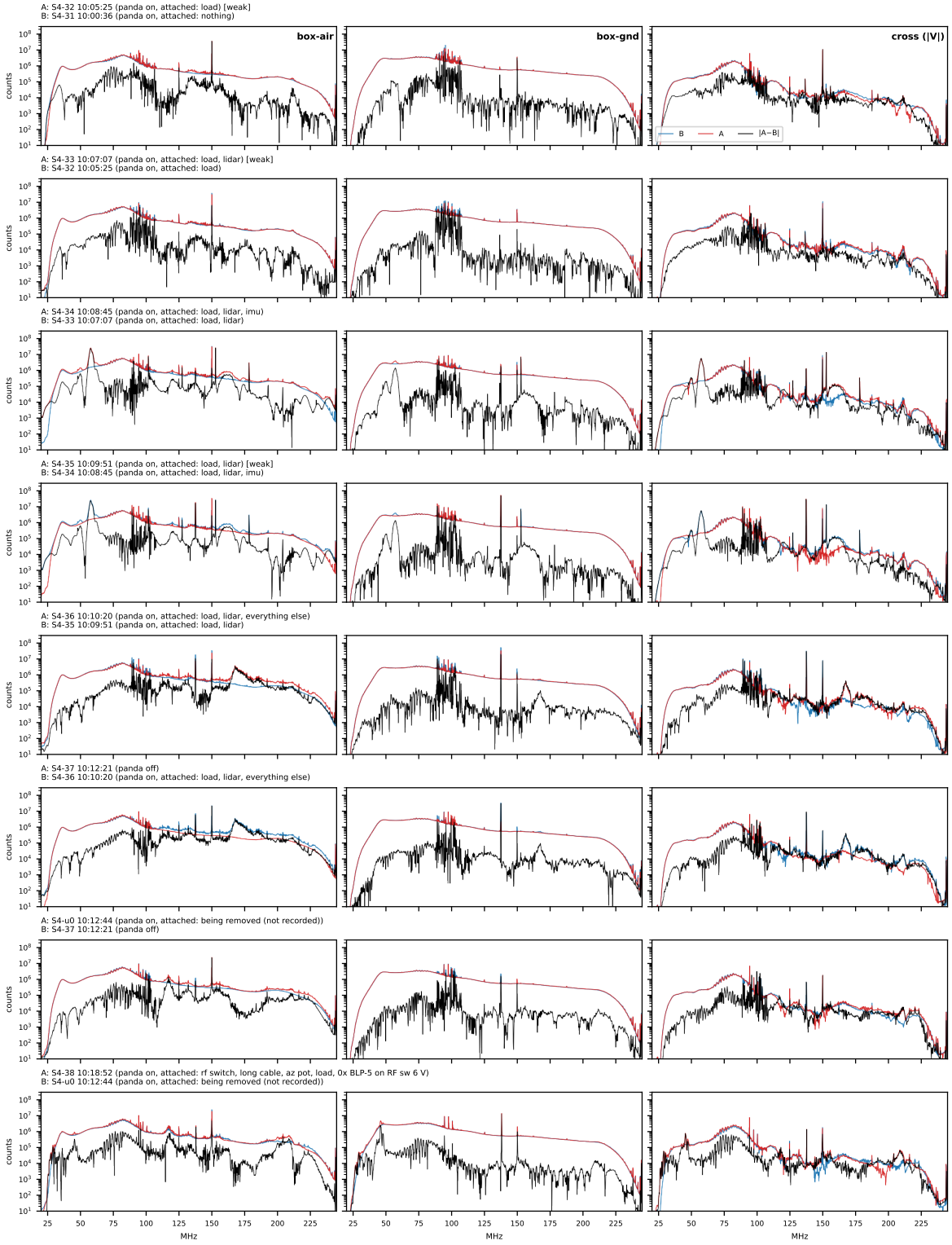


Figure 13: Session 4: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 1 of 4)

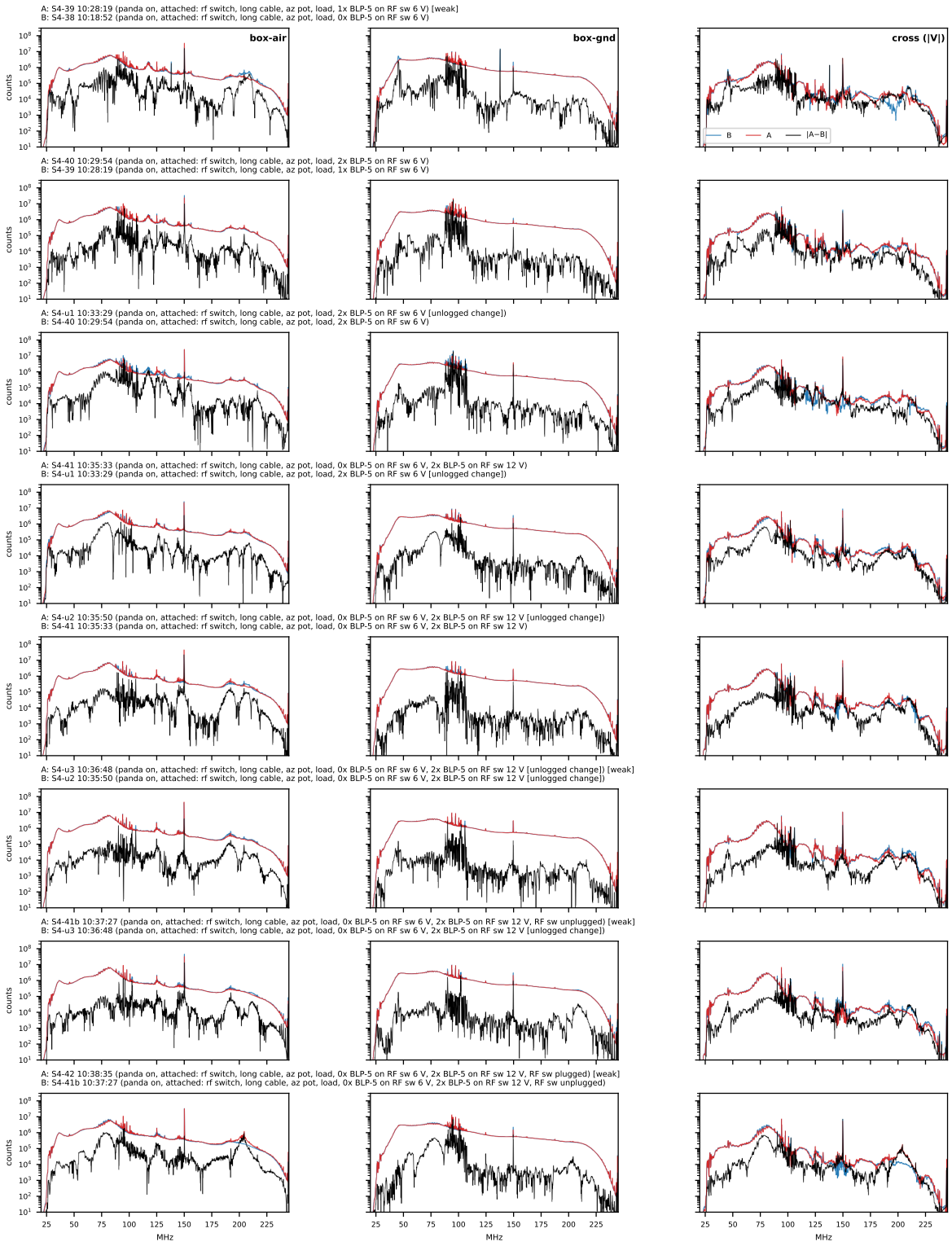


Figure 14: Session 4: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 2 of 4)

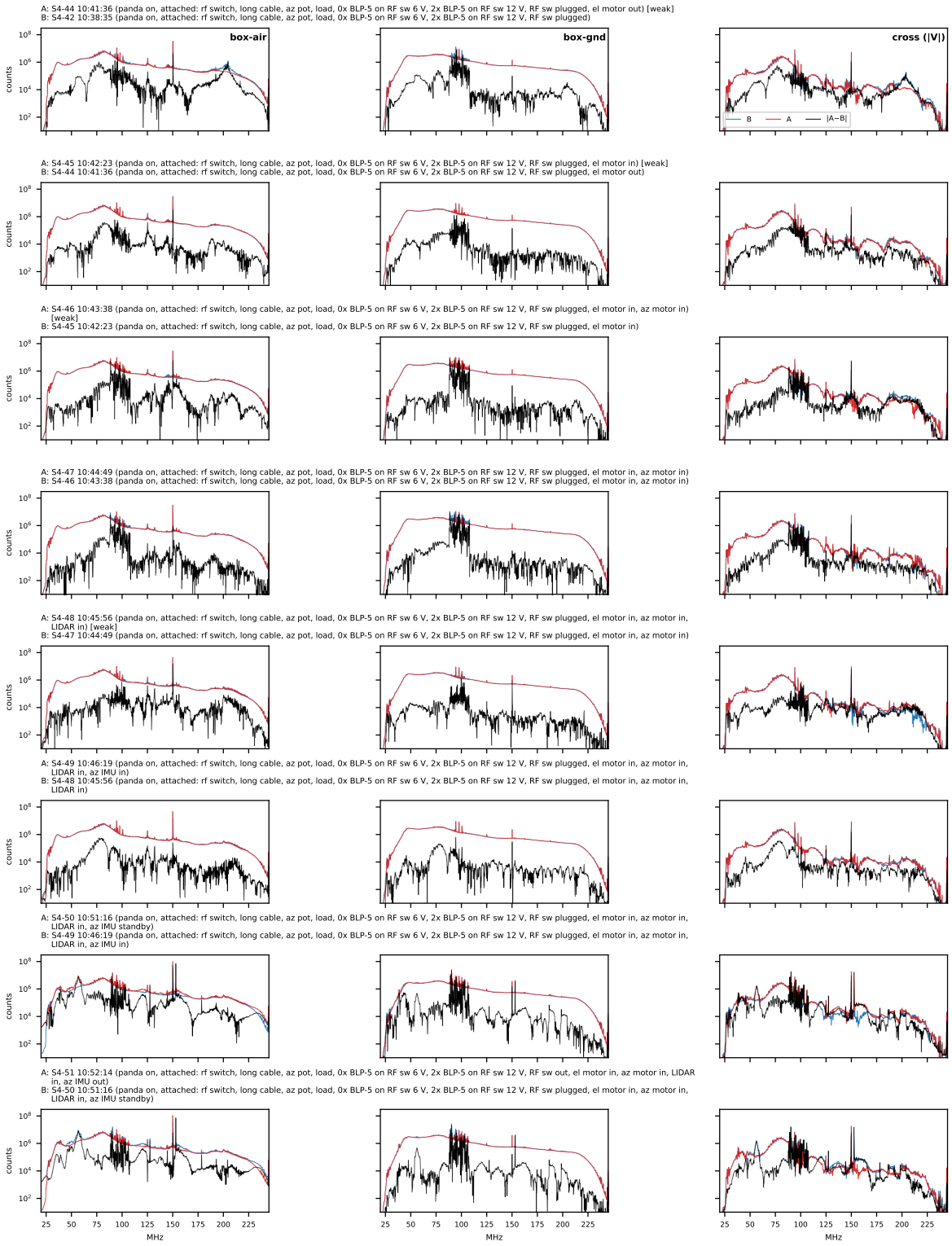


Figure 15: Session 4: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 3 of 4)

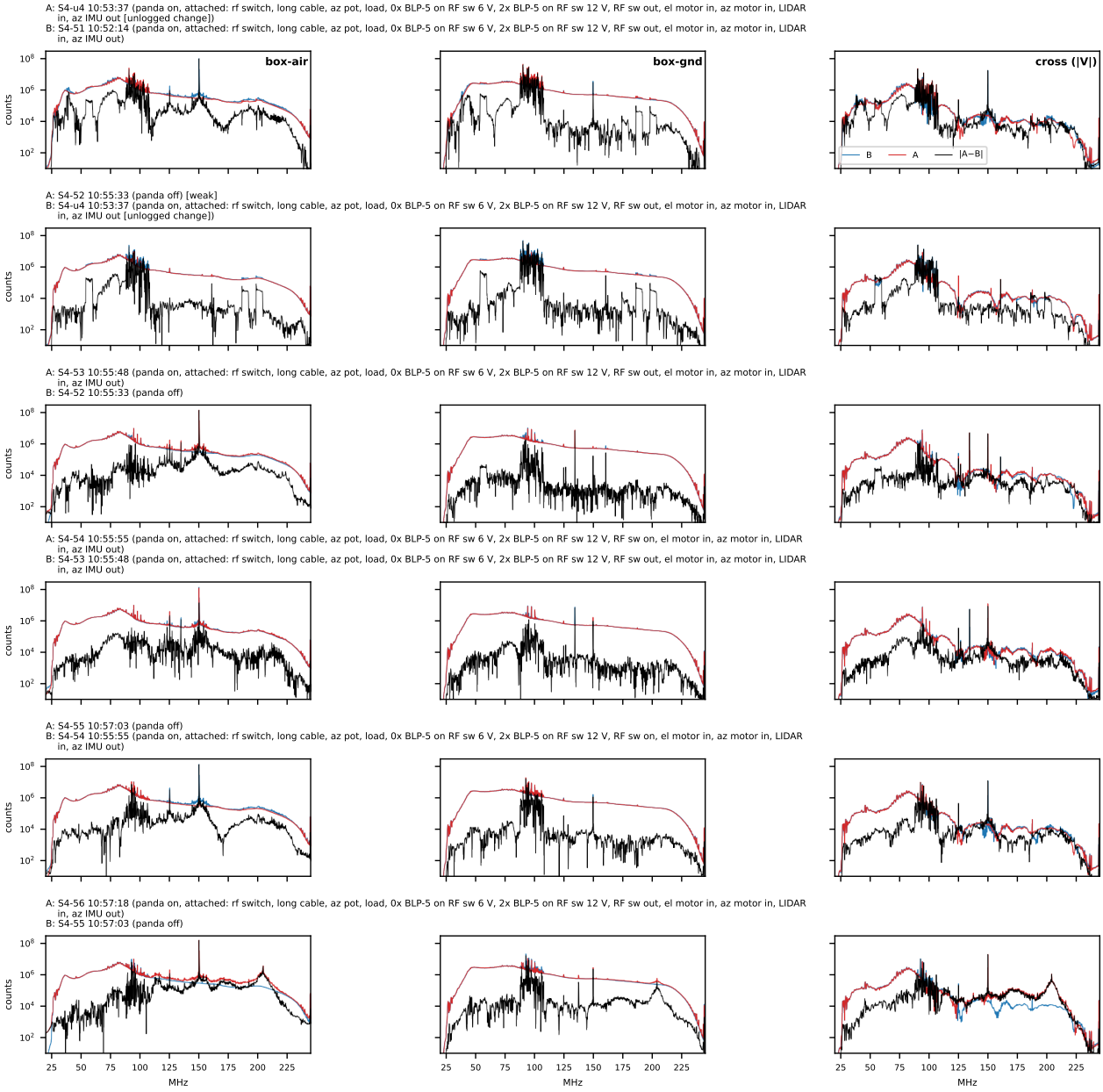


Figure 16: Session 4: A (this step, red), B (the previous step, blue) and |A−B| (black) for box-air, box-gnd and the cross ($|V|$), in counts per integration. "(weak)": no clear change at the change time. (page 4 of 4)

5 First look: the Panda's imprint

The Panda power cycles are the cleanest comparison in the data, because an off state is minutes away from an on state of the same configuration, on either side. Fig. 17 and Table 2 compare each off interval with the on intervals before and after it. With the Panda on, box-air gains 21.1–228.8% of its Panda-off power, averaged over 150–225 MHz, in every comparison. The excess carries narrow features, the clearest near 150, 205 and 240 MHz. The largest excesses (S4-36, S4-56) are on states with other equipment attached, so they include more than the Panda. The excess is in the cross too, so some of it reaches box-gnd's antenna or signal path coherently. Box-gnd's own auto-correlation changes by at most 2.6×10^4 counts over 150–225 MHz, and in session 1 by at most 6×10^3, against box-air's $(18–41) \times 10^3$. Which of the Panda's circuits make which part of this, and by which path, is the next version's question.

6 Open items

- **Interpretation.** Identify each component's spectral signature from the per-step spectra in `tables/segments.npz`, and say for each whether it reaches box-gnd (the cross) or only box-air. Physical tests are on hand: the Panda power cycles, the plug and unplug steps timed by metadata, and the filter changes.
- **"cu".** The 6 V RF battery or copper tape (S3-24, S3-27); neither step changes anything above the controls, which favours copper tape unless the RF chain had another supply.
- **Box-gnd in phase B.** Input 3's notch at 110–150 MHz moves with box-air's power configuration in session 1. Whether that is coupling or coincidence is not settled.
- **Uncertain steps.** The 22 weak steps, session 2's software steps and S4-52 have uncertain boundaries; their spectra should be used with that in mind.
- **Screenshots not used here**: 07-12 and 07-13 (shakedown, before the notes; not Aaron's captures, so their clock is unchecked and may not be PDT), 07-15 16:15 (no data: box-air outage, memo 001), 07-16 07:46 (the 1.000 MHz self-EMI, memo 006) and 07-17 (the transmitter, used only for the clock).

A Every step

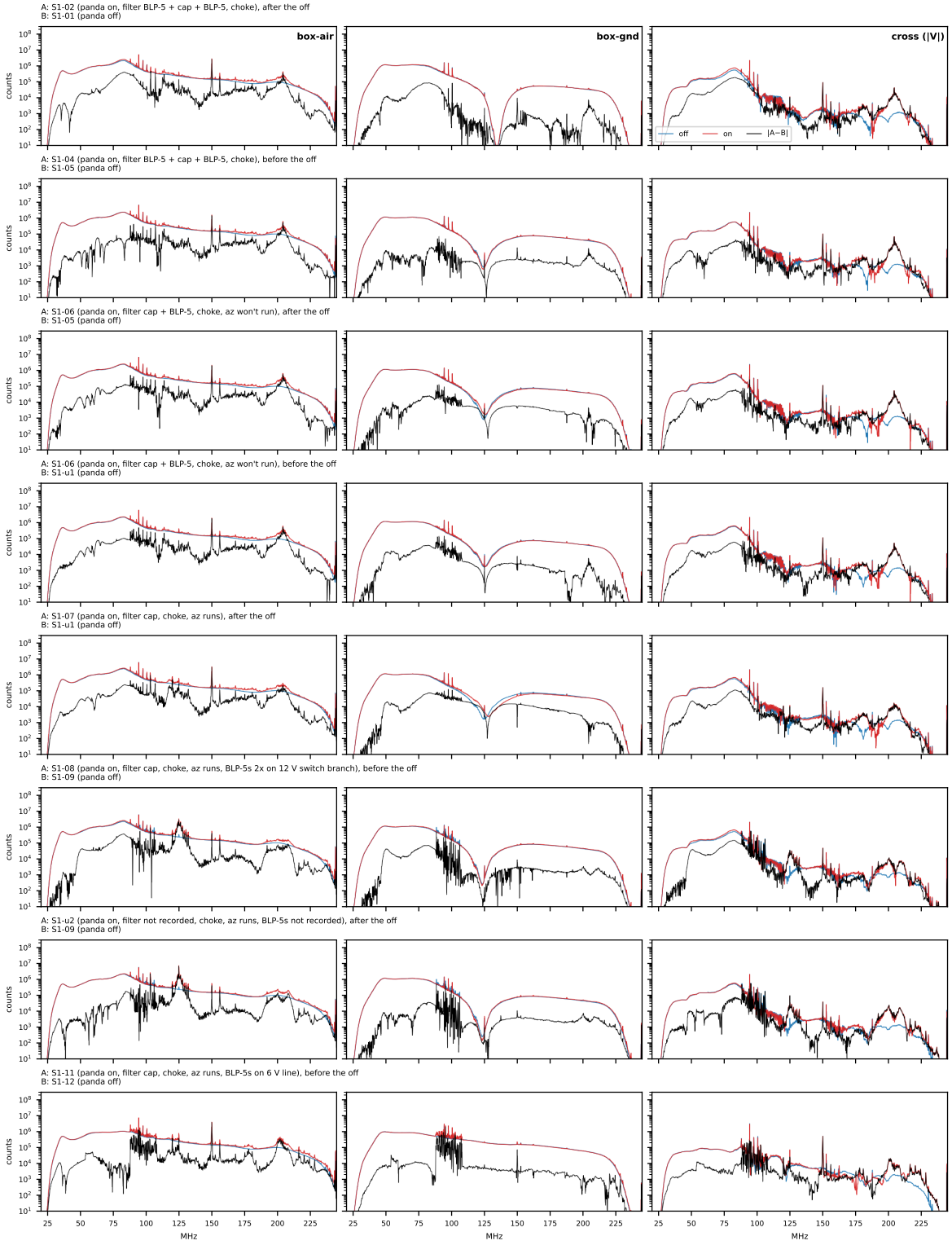


Figure 17: Panda on (A, red) against Panda off (B, blue), and |A−B| (black), for every power cycle; "before"/"after" says which side of the power-off the on state is. (page 1 of 2)

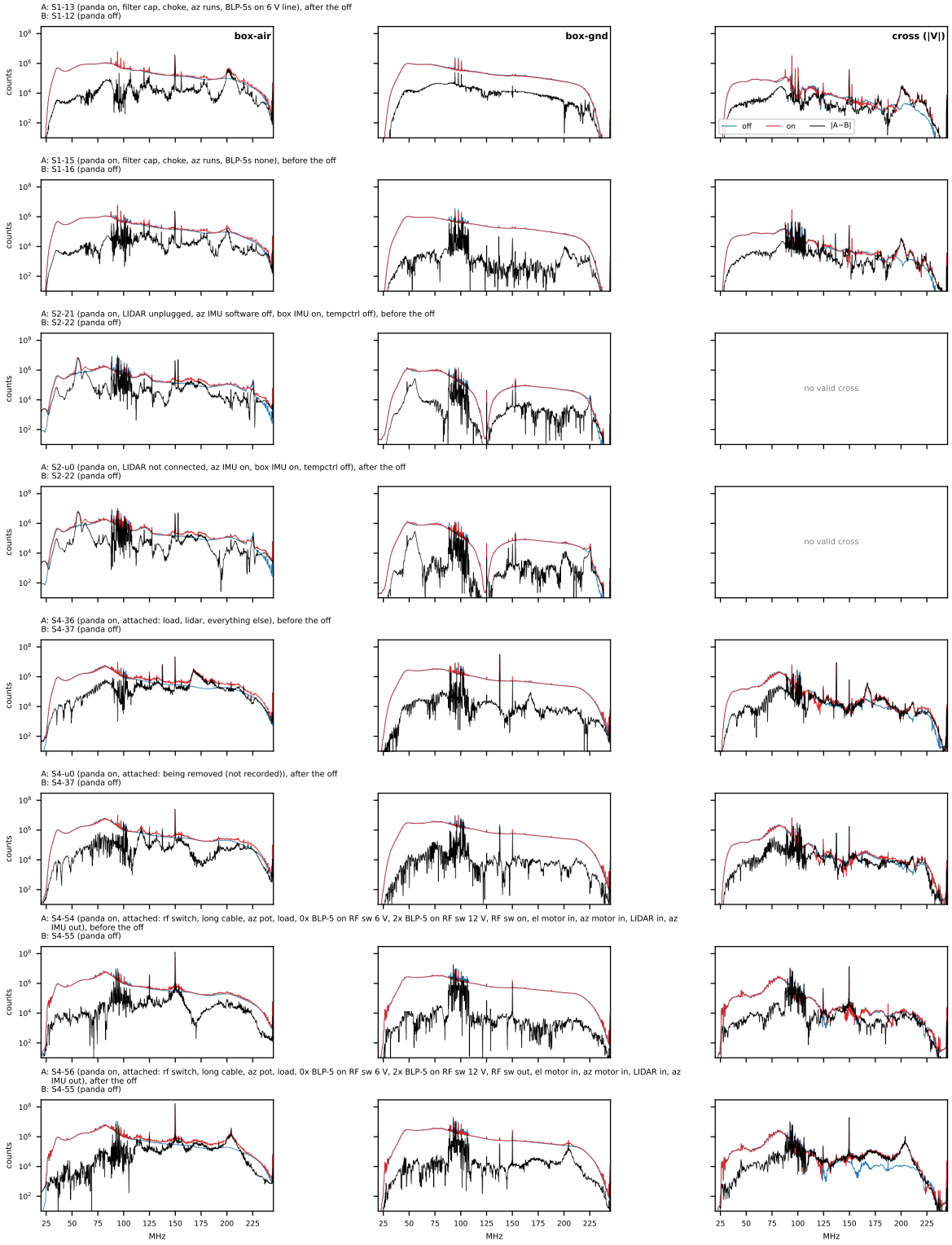


Figure 18: Panda on (A, red) against Panda off (B, blue), and |A−B| (black), for every power cycle; "before"/"after" says which side of the power-off the on state is. (page 2 of 2)

Table 2: Panda on against Panda off: band means of on−off for box-air and box-gnd and of |on−off| for the cross (10^3 counts per integration; session 4's integrations are twice as long), and the percentage of channels where box-air or the cross changes by more than three times the null (§3.3).

Panda on	off	on state	box-air on−off		box-gnd	|cross|	box-air	cross
			30–85	150–225	150–225	150–225	% > 3×	% > 3×
			(band mean, 10^3 counts)				null	null
S1-02	S1-01	after	+69	+34	+0	2	45	21
S1-04	S1-05	before	-4	+34	+2	3	64	35
S1-06	S1-05	after	+31	+39	-2	3	63	32
S1-06	S1-u1	before	+33	+39	-1	3	69	33
S1-07	S1-u1	after	+44	+33	-5	2	70	28
S1-08	S1-09	before	+68	+18	+2	2	69	34
S1-u2	S1-09	after	+7	+24	+3	3	59	21
S1-11	S1-12	before	+11	+38	-2	3	51	12
S1-13	S1-12	after	+8	+41	-6	3	71	32
S1-15	S1-16	before	+1	+22	+1	3	62	30
S2-21	S2-22	before	+476	+66	+1	–	75	–
S2-u0	S2-22	after	+437	+66	+2	–	77	–
S4-36	S4-37	before	-98	+407	+9	28	48	7
S4-u0	S4-37	after	-62	+63	+6	8	14	2
S4-54	S4-55	before	+0	+60	+2	6	54	32
S4-56	S4-55	after	-14	+326	+26	75	12	2

Table 3: Every step: its screenshot and change time (PDT), how the change was found (fit = spectral fit; meta = sensor metadata; eye = by eye; drop = end of a box-air dropout; italic: weak, statistic $< 4\times$ its local median), the averaging interval and integrations kept, the correlator file (`corr_<file>.h5`) and integration index of the first integration after the change, the configuration after the step (restated from the notes, cumulative within a session), and the field note verbatim.

step	shot	change	src	int. s	N	file	idx	configuration	field note
S1: Panda power-input filtering (2026-07-14)									
01	08:47:07	08:46:21	fit	42	156	20260714_154814Z	98	panda off	panda off
02	08:48:59	08:47:08	fit	108	402	20260714_154815Z	31	panda on, filter BLP-5 + cap + BLP-5, choke	panda on, blp-5 + cap + blp-5, choke
03	08:49:58	08:48:59	fit	38	142	20260714_154914Z	207	panda on, filter BLP-5 + cap + BLP-5, no choke	panda on, blp-5 + cap + blp-5
04	08:50:32	*08:49:41*	fit	423	1578	20260714_155018Z	123	panda on, filter BLP-5 + cap + BLP-5, choke	panda on, blp-5 + cap + blp-5, choke
05	08:57:15	08:56:49	fit	15	58	20260714_155750Z	35	panda off	panda off
06	08:57:44	08:57:08	fit	328	1224	20260714_155750Z	107	panda on, filter cap + BLP-5, choke, az won't run	panda on, cap + blp-5, choke, az doesn't run
u1		09:02:41	eye	20	75	20260714_160340Z	146	panda off	[unlogged] Panda off (unlogged): 150-225 MHz excess vanishes in box-air and cross; boot flash 09:03:06
07	09:03:36	09:03:04	fit	395	1472	20260714_160340Z	235	panda on, filter cap, choke, az runs	panda on, cap, choke, az runs
08	09:10:52	09:09:44	fit	74	255	20260714_161042Z	43	panda on, filter cap, choke, az runs, BLP-5s 2x on 12 V switch branch	panda on, cap, choke, 2x blp-5 on 12V-switch branch
09	09:11:23	*09:11:02*	eye	11	41	20260714_161201Z	92	panda off	panda off
u2		09:11:17	eye	217	793	20260714_161201Z	148	panda on, filter not recorded, choke, az runs, BLP-5s not recorded	[unlogged] Panda back on after the boot flash at 09:11:16 (unlogged); configuration until 09:14:58 not recorded
10	09:15:37	09:14:58	eye	447	796	20260714_161605Z	11	panda on, filter cap, choke, az runs, BLP-5s none	panda on, cap, choke, no blp-5s
11	09:25:30	09:22:29	drop	205	764	20260714_162336Z	12	panda on, filter cap, choke, az runs, BLP-5s on 6 V line	panda on, cap, choke, blp-5s on 6V line
12	09:26:17	09:25:58	fit	66	246	20260714_162749Z	71	panda off	panda off
13	09:27:36	09:27:08	fit	195	726	20260714_162753Z	91	panda on, filter cap, choke, az runs, BLP-5s on 6 V line	panda on, cap, choke, blp-5s on 6V line
14	09:31:29	09:30:26	fit	117	231	20260714_163106Z	111	panda on, filter cap, choke, az runs, BLP-5s on 6 V line	panda on, cap, choke, blp-5s on 6V line
15	09:32:47	09:32:27	drop	372	1385	20260714_163315Z	81	panda on, filter cap, choke, az runs, BLP-5s none	panda on, cap, choke, no blp-5s
16	09:39:11	09:38:43	fit	49	182	20260714_164225Z	41	panda off	panda off
S2: Sensors: LIDAR, IMUs, temperature controller (2026-07-14)									
01	15:50:37	15:50:16	drop	33	84	20260714_225036Z	164	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	All on except tempctrl module
02	15:51:17	15:51:02	meta	19	59	20260714_225141Z	93	panda on, LIDAR unplugged, az IMU on, box IMU on, tempctrl off	Physical disconnect lidar
03	15:52:06	*15:51:51*	meta	17	52	20260714_225245Z	35	panda on, LIDAR software off, az IMU on, box IMU on, tempctrl off	Software disconnect lidar
04	15:52:35	15:52:16	meta	18	57	20260714_225245Z	128	panda on, LIDAR on, az IMU unplugged, box IMU on, tempctrl off	physical disconnect IMU
05	15:53:13	*15:52:56*	meta	40	129	20260714_225350Z	37	panda on, LIDAR on, az IMU software off, box IMU on, tempctrl off	software disconnect IMU
06	15:54:36	*15:54:12*	meta	35	95	20260714_225454Z	81	panda on, LIDAR on, az IMU on, box IMU on, tempctrl on	Everything on (including tempctrl)
07	15:55:46	15:55:30	meta	58	208	20260714_225558Z	131	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	Same as 1.
08	15:56:47	*15:56:32*	meta	22	82	20260714_225703Z	122	panda on, LIDAR on, az IMU on, box IMU software off, tempctrl off	Box (el) IMU software disconnect
09		15:56:58	meta	48	133	20260714_225703Z	219	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	Same as 1.
10	16:00:39	15:58:30	meta	234	849	20260714_225912Z	82	panda on, LIDAR software off, az IMU on, box IMU on, tempctrl off	Lidar off via software
11	16:02:45	16:02:28	meta	29	105	20260714_230329Z	8	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	same as 1.
12	16:03:18	16:03:01	meta	–	–	20260714_230329Z	131	panda on, LIDAR on, az IMU software off, box IMU on, tempctrl off	Turn off Az IMU software
13	16:03:34	*16:03:21*	meta	28	103	20260714_230329Z	206	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	Turn back on (same as 1)
14	16:04:14	16:04:00	meta	17	48	20260714_230434Z	111	panda on, LIDAR on, az IMU unplugged, box IMU on, tempctrl off	Physical Imu disconnect
15	16:04:34	16:04:21	meta	19	72	20260714_230434Z	189	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	Plug back in IMU
17	16:05:06	16:04:54	meta	8	21	20260714_230538Z	72	panda on, LIDAR unplugged, az IMU on, box IMU on, tempctrl off	Physical lidar disconnect
18	16:05:26	*16:05:11*	meta	55	176	20260714_230538Z	136	panda on, LIDAR unplugged, az IMU unplugged, box IMU on, tempctrl off	Physical lidar + IMU disconnect
19	16:06:24	16:06:10	meta	12	43	20260714_230643Z	115	panda on, LIDAR unplugged, az IMU on, box IMU on, tempctrl off	IMU reconnect + lidar disconnect physical
20	16:06:43	16:06:31	meta	71	266	20260714_230643Z	194	panda on, LIDAR unplugged, az IMU software off, box IMU on, tempctrl off	IMU disconnect in software
21	16:08:20	16:07:49	fit	81	298	20260714_230852Z	7	panda on, LIDAR unplugged, az IMU software off, box IMU on, tempctrl off	Repeat of 20.
22	16:09:39	16:09:24	meta	40	134	20260714_231046Z	118	panda off	Panda powered off.

step	shot	change	src	int. s	N	file	idx	configuration	field note
u0		16:10:08	eye	61	200	20260714_231100Z	42	panda on, LIDAR not connected, az IMU on, box IMU on, tempctrl off	[unlogged] Panda booted (flash 16:10:06), LIDAR not yet connected (unlogged)
23	16:11:33	16:11:13	eye	86	294	20260714_231205Z	44	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off	panda on + lidar connected
24	16:12:55	16:12:43	meta	104	222	20260714_231309Z	139	panda on, LIDAR unplugged, az IMU unplugged, box IMU on, tempctrl off	physical disconnect imu + lidar
25	16:14:46	16:14:34	drop	167	617	20260714_231518Z	74	panda on, LIDAR on, az IMU on, box IMU on, tempctrl off, load off	connecting all except load
u1	16:17:49	16:17:25	meta	39	147	20260714_231727Z	230	panda on, LIDAR not answering, az IMU on, box IMU not answering, tempctrl off, load off	[unlogged] imu_el error; lidar error from 16:17:31
S3: Sensors, load, CU and azimuth drive (2026-07-14)									
18	19:20:20	19:19:15	fit	73	274	20260715_021939Z	154	panda on, az IMU unplugged, LIDAR unplugged, load unplugged	panda on, imu unplug, lidar unplug, load unplug
19a	19:24:48	19:20:32	fit	292	1088	20260715_022044Z	202	panda on, az IMU unplugged, LIDAR unplugged, load unplugged	panda on, imu unplug, lidar unplug, load unplug
19b	19:25:55	19:25:29	meta	9	34	20260715_022606Z	105	panda on, az IMU plugged, LIDAR unplugged, load unplugged	panda on, imu plug, lidar unplug, load unplug, (christian)
19c	19:27:57	19:25:42	fit	147	547	20260715_022606Z	154	panda on, az IMU standby, LIDAR unplugged, load unplugged	panda on, imu standby, lidar unplug, load unplug
20	19:28:32	19:28:13	meta	83	311	20260715_022815Z	236	panda on, az IMU unplugged, LIDAR unplugged, load unplugged	panda on, imu unplug, lidar unplug, load unplug
21	19:30:04	19:29:43	eye	32	120	20260715_023024Z	91	panda on, az IMU unplugged, LIDAR plugged, load unplugged	19-30a, panda on, imu unplug, lidar plug, load unplug
22	19:30:44	*19:30:19*	fit	45	170	20260715_023024Z	226	panda on, az IMU unplugged, LIDAR standby, load unplugged	19-30b, panda on, imu unplug, lidar standby, load unplug
23	19:31:21	19:31:08	fit	66	246	20260715_023128Z	170	panda on, az IMU unplugged, LIDAR unplugged, load unplugged	panda on, imu unplug, lidar unplug, load unplug
24	19:33:47	19:32:18	fit	174	647	20260715_023232Z	190	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu off	panda on, imu unplug, lidar unplug, load unplug cu off
25	19:35:49	19:35:16	meta	85	318	20260715_023546Z	132	panda on, az IMU plugged, LIDAR unplugged, load plugged, cu off	19-34, panda on, imu plug, lidar unplug, load plug
26	19:37:17	*19:36:46*	meta	88	330	20260715_023650Z	227	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu off	panda on, imu unplug, lidar unplug, load unplug
27	19:38:37	19:38:21	fit	208	774	20260715_023859Z	102	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu on	panda on, imu unplug, lidar unplug, load unplug cu on
28	19:44:18	19:41:52	fit	142	530	20260715_024212Z	170	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu on, az stopped	19-44a, panda on, imu unplug, lidar unplug, load unplug cu on, az stopped
29	19:44:29	19:44:18	fit	6	22	20260715_024421Z	234	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu on, az moving	19-44b, ... az moving
30	19:44:48	19:44:28	fit	40	151	20260715_024526Z	30	panda on, az IMU unplugged, LIDAR unplugged, load unplugged, cu on, az stopped	19-44c, ... az stopped
S4: Reattaching sensors, RF switch, motors (2026-07-15)									
31	10:01:30	10:00:36	fit	256	476	20260715_170239Z	10	panda on, attached: nothing	panda on, nothing attached
32	10:06:12	*10:05:25*	meta	99	184	20260715_170657Z	67	panda on, attached: load	panda on, load on
33	10:08:21	*10:07:07*	fit	93	172	20260715_170906Z	18	panda on, attached: load, lidar	panda on, load on, lidar on
34	10:09:14	10:08:45	meta	62	115	20260715_170906Z	199	panda on, attached: load, lidar, imu	panda on, load on, lidar on, imu plugged in
35	10:10:09	*10:09:51*	meta	26	48	20260715_171115Z	82	panda on, attached: load, lidar	panda on, load on, lidar on, imu unplugged
36	10:12:14	10:10:20	fit	116	215	20260715_171115Z	138	panda on, attached: load, lidar, everything else	10-12-09, panda on, load on, lidar on, imu unplugged, everything else plugged
37	10:12:34	10:12:21	meta	19	35	20260715_171323Z	122	panda off	panda off
u0		10:12:44	eye	364	675	20260715_171323Z	164	panda on, attached: being removed (not recorded)	[unlogged] Panda back on (boot flash 10:12:42; unlogged); sensors removed during 10:12:44-10:18 without notes
38	10:25:03	10:18:52	fit	563	904	20260715_171950Z	130	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V	panda on, rf sw on, long cable on, az pot on, load on. think that rf switch w/o blp-5 filters is a problem
39	10:28:46	*10:28:19*	drop	91	91	20260715_172825Z	226	panda on, attached: rf switch, long cable, az pot, load, 1x BLP-5 on RF sw 6 V	panda on, rf sw on with 1x blp-5 on 6V line, long cable on, az pot on, load on
40	10:30:14	10:29:54	drop	211	252	20260715_173034Z	163	panda on, attached: rf switch, long cable, az pot, load, 2x BLP-5 on RF sw 6 V	(logged as 10-28-46) panda on, rf sw on with 2x blp-5 on 6V line, long cable on, az pot on, load on
u1	10:33:51	10:33:29	drop	120	167	20260715_173452Z	84	panda on, attached: rf switch, long cable, az pot, load, 2x BLP-5 on RF sw 6 V [unlogged change]	[unlogged]
41	10:36:07	10:35:33	drop	14	18	20260715_173701Z	74	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V	panda on, rf sw on with 0x blp-5 on 6V line, 2 on 12V, long cable on, az pot on, load on

step	shot	change	src	int. s	N	file	idx	configuration	field note
u2	10:37:04	10:35:50	fit	54	100	20260715_173701Z	107	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V [unlogged change]	(after 'skip some')
u3	10:37:17	*10:36:48*	fit	35	64	20260715_173701Z	215	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V [unlogged change]	[unlogged]
41b	10:38:38	*10:37:27*	fit	65	120	20260715_173910Z	47	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw unplugged	10-38-38-38, panda on, rf sw on with 0x blp-5 on 6V line rfsw unplugged
42	10:38:43	*10:38:35*	fit	176	328	20260715_173910Z	175	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged	10-38-38-43, panda on, rf sw on with 0x blp-5 on 6V line rfsw plugged (43 repeats this line)
44	10:42:26	*10:41:36*	fit	43	80	20260715_174327Z	31	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor out	el motor out
45	10:42:41	*10:42:23*	fit	71	132	20260715_174327Z	119	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in	el motor in
46	10:43:51	*10:43:38*	fit	67	124	20260715_174536Z	19	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in, az motor in	el motor in, az motor in
47	10:45:52	10:44:49	fit	63	116	20260715_174536Z	151	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in, az motor in	same, just later
48	10:46:20	*10:45:56*	fit	20	36	20260715_174745Z	35	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in, az motor in, LIDAR in	el motor in, az motor in, lidar in
49	10:51:36	10:46:19	fit	292	544	20260715_174745Z	79	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in, az motor in, LIDAR in, az IMU in	el motor in, az motor in, lidar in, imu in
50	10:52:14	10:51:16	fit	54	100	20260715_175203Z	151	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw plugged, el motor in, az motor in, LIDAR in, az IMU standby	el motor in, az motor in, lidar in, imu standby
51	10:52:23	10:52:14	fit	79	147	20260715_175622Z	19	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw out, el motor in, az motor in, LIDAR in, az IMU out	el motor in, az motor in, lidar in, imu out, rfsw out
u4		10:53:37	eye	112	209	20260715_175622Z	174	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw out, el motor in, az motor in, LIDAR in, az IMU out [unlogged change]	[unlogged] unlogged change: box-air 150-225 MHz excess falls by ∼1 dB
52	10:55:49	*10:55:33*	eye	11	21	20260715_175622Z-1	150	panda off	panda off
53	10:55:59	10:55:48	eye	3	4	20260715_175622Z-1	178	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw out, el motor in, az motor in, LIDAR in, az IMU out	el motor in, az motor in, lidar in, imu out, rfsw out
54	10:57:03	10:55:55	fit	65	120	20260715_175622Z-1	191	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw on, el motor in, az motor in, LIDAR in, az IMU out	el motor in, az motor in, lidar in, imu out, rfsw on
55	10:57:12	10:57:03	fit	11	20	20260715_175829Z	79	panda off	panda off
56	11:04:03	10:57:18	fit	425	787	20260715_175829Z	107	panda on, attached: rf switch, long cable, az pot, load, 0x BLP-5 on RF sw 6 V, 2x BLP-5 on RF sw 12 V, RF sw out, el motor in, az motor in, LIDAR in, az IMU out	lidar in, rfsw out

B The clock and the metadata-timed steps

Read as site time (MDT, UTC−6), the 07-14 notes fall an hour before the tests: input 3 is dead at 08:43–09:35 MDT, while the screenshots of 08:47–09:39 show it live. The 07-17 transmitter on/off test fixes the offset (Fig. 19). Box-gnd sees the transmitter's teeth vanish from 18:44:14 to 18:44:21 UTC; three screenshots at 11:44:19, :31 and :41 on the laptop clock show it on, off, on. Under UTC−7 they are 18:44:19, :31, :41 UTC and show those states if the dashboard displays data 8.6–16.6 s old; under UTC−6 they would be 17:44 UTC, when the transmitter was off for 51 minutes (memo 001). On 07-14 and 07-15 every step timed by a sensor's own metadata precedes its screenshot by 12.0 s or more, consistent with the lag. Table 4 lists those steps.

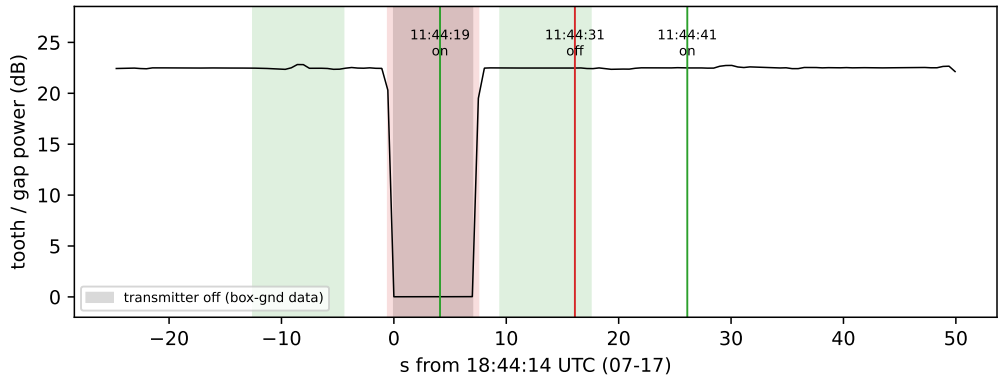


Figure 19: Transmitter tooth contrast seen by box-gnd through the 07-17 on/off test; grey, its off interval in the data. Vertical lines: the three screenshots at UTC=laptop+7 h, labelled with the state they show; coloured bands, where each screenshot's displayed data lie for the lag bounds.

Table 4: Steps timed by sensor metadata. $S/\tilde{S}$: the step statistic at that time over its local median (below 4: no clear spectral change).

step	sensor change (metadata)	time (PDT)	$S/\tilde{S}$
S2-02	lidar status turns to error	15:51:02	18.5
S2-03	lidar error again (it had reported from 15:51:25)	15:51:51	3.1
S2-04	imu_az error	15:52:16	10.9
S2-05	imu_az error again (reported from 15:52:42)	15:52:56	1.7
S2-06	tempctrl_load starts reporting	15:54:12	1.9
S2-07	tempctrl_load stops reporting	15:55:30	16.9
S2-08	imu_el error	15:56:32	0.4
S2-09	imu_el reports again	15:56:58	8.7
S2-10	lidar error, continuous to 16:02:28	15:58:30	11.7
S2-11	lidar reports again	16:02:28	19.0
S2-12	imu_az error	16:03:01	9.7
S2-13	imu_az reports again	16:03:21	0.2
S2-14	imu_az error	16:04:00	6.5
S2-15	imu_az reports again	16:04:21	5.4
S2-17	lidar error	16:04:54	13.5
S2-18	imu_az error	16:05:11	2.9
S2-19	imu_az reports	16:06:10	15.5
S2-20	imu_az error	16:06:31	4.2
S2-22	all metadata stops (box-air 150-225 MHz excess falls at 16:0	16:09:24	17.2
S2-24	imu_az and lidar error	16:12:43	6.6
S2-u1	imu_el error; lidar error from 16:17:31	16:17:25	9.1
S3-19b	imu_az starts reporting	19:25:29	8.9
S3-20	imu_az stops reporting	19:28:13	14.3
S3-25	tempctrl_load starts reporting	19:35:16	7.9
S3-26	tempctrl_load stops reporting	19:36:46	3.9
S4-32	tempctrl_load starts reporting	10:05:25	2.4
S4-34	imu_az starts reporting	10:08:45	6.3
S4-35	imu_az stops reporting	10:09:51	3.6
S4-37	all metadata but potmon stops	10:12:21	4.2

C Reproducing this memo

`python make_products.py && pdflatex memo-007.tex && pdflatex memo-007.tex` in this directory. It reads `tests.json` and the raw correlator files of `marjum-2026-07/data/` (set `EIGSEP_CAMPAIGN_ROOT` from a worktree), records the SHA-256 of each in `tables/inputs.json`, and takes about a minute. The change-time search and the averaging are `process_session`; the constants (lag bounds, guard, thresholds) are at the top of the script.

gitignored input	SHA-256 (first 16)
`data/` (158 files; digest of their SHA-256s)	`1da051c404062759`
eigsep commit	`8347e78-dirty`
generated	2026-10-04 16:23 UTC

Revision history

- 2026-10-03: first version: the four sessions documented, the screenshot clock established, per-step spectra extracted.
- 2026-10-04: "cu" read as the 6 V RF battery or copper tape, with the data's constraint. Changes shown and summarized as additive differences (A, B, |A−B|) instead of ratios; time alignment moved to an appendix.